

Strategy Profile: FIFO

Isolated Performance and Tail-Risk Analytics

GENERATED	MONTE CARLO RUNS	STRATEGIES TESTED
10 May 2026 16:41	40	2
METHOD	UNIT	REPORT TYPE
Discrete-Event Simulation (DES)	Haemodialysis Centre	Comparative Performance Analysis

Abstract

This report presents the results of a Monte Carlo discrete-event simulation study evaluating scheduling strategies for a haemodialysis unit. Stochastic variability in patient volume, machine availability, session duration, and nursing resources is modelled across 40 independent runs per strategy. Distributional summaries, tail-risk metrics (p90 / p99), and head-to-head strategy comparisons are provided to support operational decision-making.

Keywords: Monte Carlo | Discrete-Event Simulation | Haemodialysis | Scheduling | Tail-Risk

1. Simulation Setup

The parameters below fully specify the stochastic environment sampled during each Monte Carlo iteration. All stochastic quantities are drawn independently per run using a seeded Mersenne Twister PRNG.

TEMPORAL BOUNDS

Parameter	Value	Unit	Description
Shift Duration	360	min	Hard wall -- 6 h total shift
Nominal Session Duration	210 - 240	min	Prescribed per-patient session (~4 h target)
Min Viable Session	180	min	Sessions shorter than this are marked failed (3 h)
Machine Cooldown	60	min	Mandatory inter-session machine rest period

RESOURCE RANGES (uniform draw per iteration)

Parameter	Value	Unit	Description
Total Machines	15 - 20	units	Physical dialysis stations available
Patient Volume	15 - 20	patients	Demand sampled each shift
Nursing Staff	2 - 4	nurses	Concurrent supervision capacity
Session Duration	210 - 240	min	Per-patient dialysis time
Machine Ready Delay	0 - 90	min	Post-cooldown preparation lag

STOCHASTIC EVENT GENERATORS

Parameter	Value	Unit	Description
Patient Arrival Offset	Uniform[0, 60]	min	Arrival jitter relative to shift start
Setup Duration	Uniform[10, 20]	min	Machine preparation time per patient

PROBABILITY METRICS

Parameter	Value	Unit	Description
Machine Defect Probability	15.0%	--	Independent Bernoulli draw per machine per session

EXPERIMENT PARAMETERS

Parameter	Value	Unit	Description
Monte Carlo Iterations	40	runs	Independent replications per strategy

2. Statistical Summary

Distributional metrics across all Monte Carlo iterations for this strategy.

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	29.1 +/- 11.7	28.2	47.1	50.9
Max Wait Time (min)	52.2 +/- 17.7	48.0	75.5	91.0
Avg Session Time (min)	224.9 +/- 2.6	224.7	228.2	230.8

3. Distribution & Utilisation

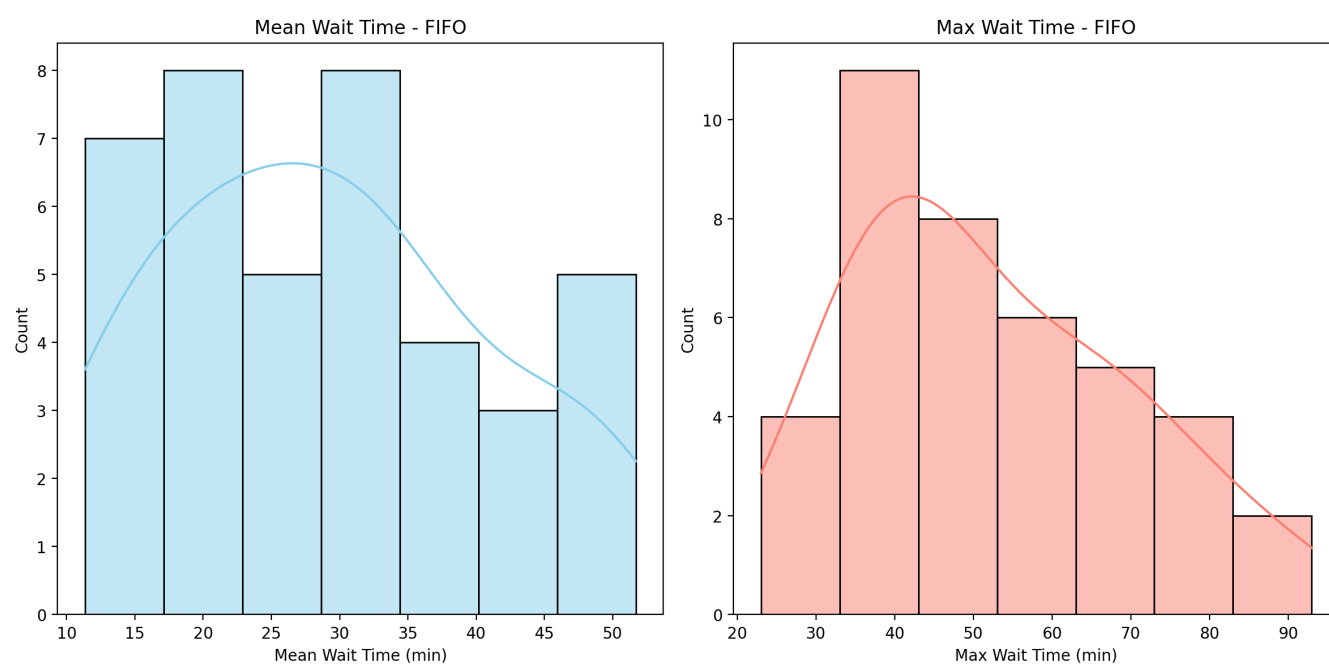


Figure 1. Wait-time distribution across Monte Carlo runs.

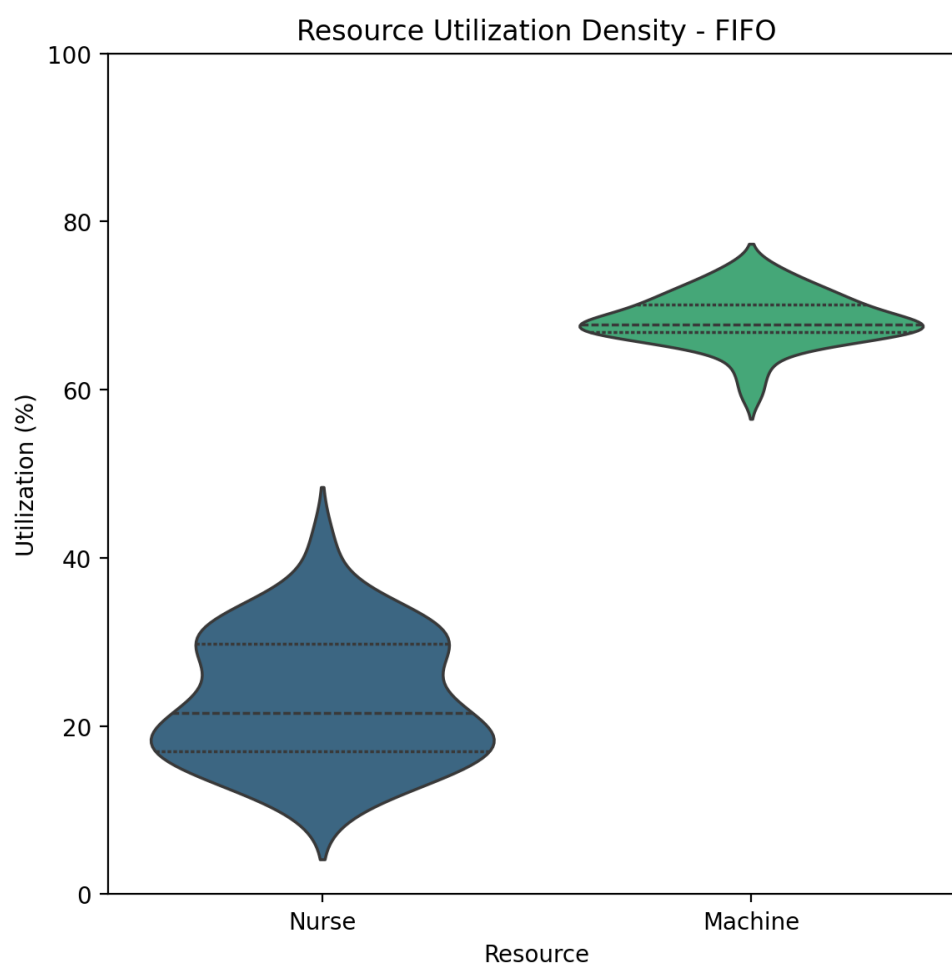


Figure 2. Machine and nurse utilisation.

4. Advanced Insights

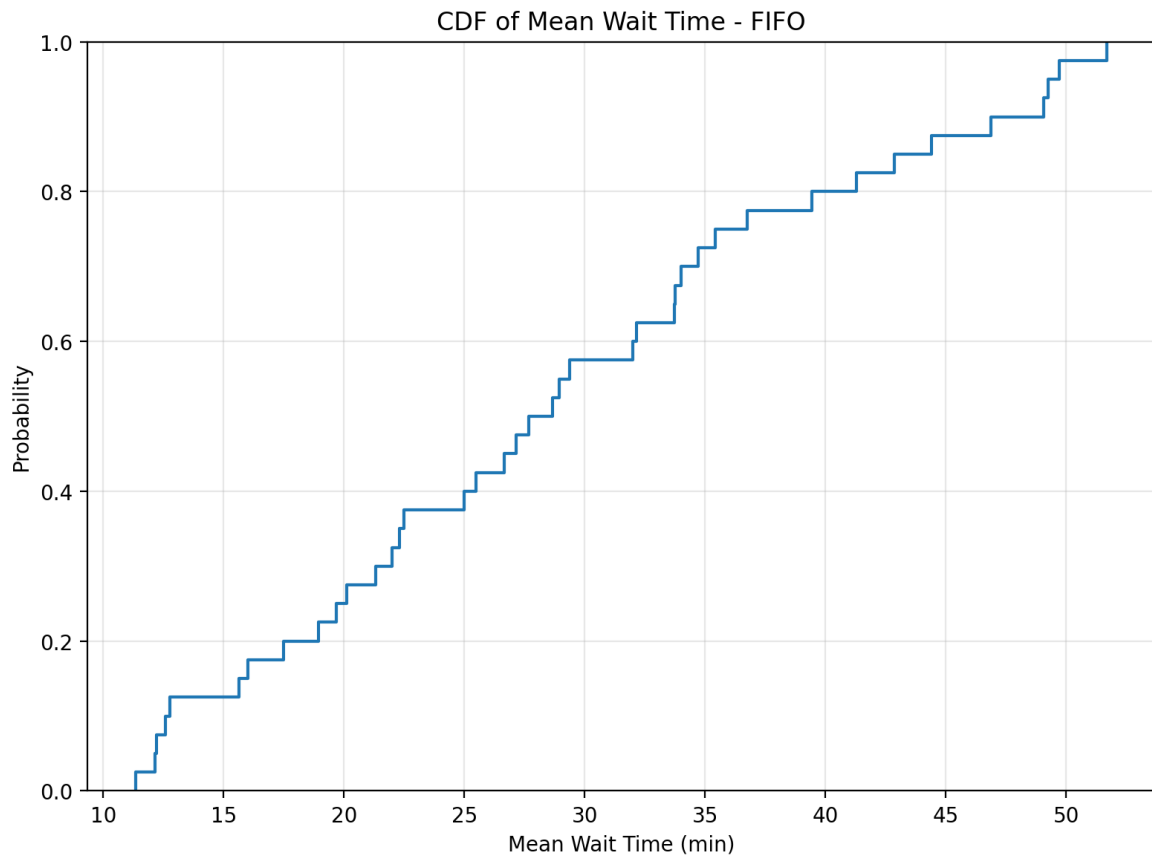


Figure 3. Empirical CDF of mean wait time.

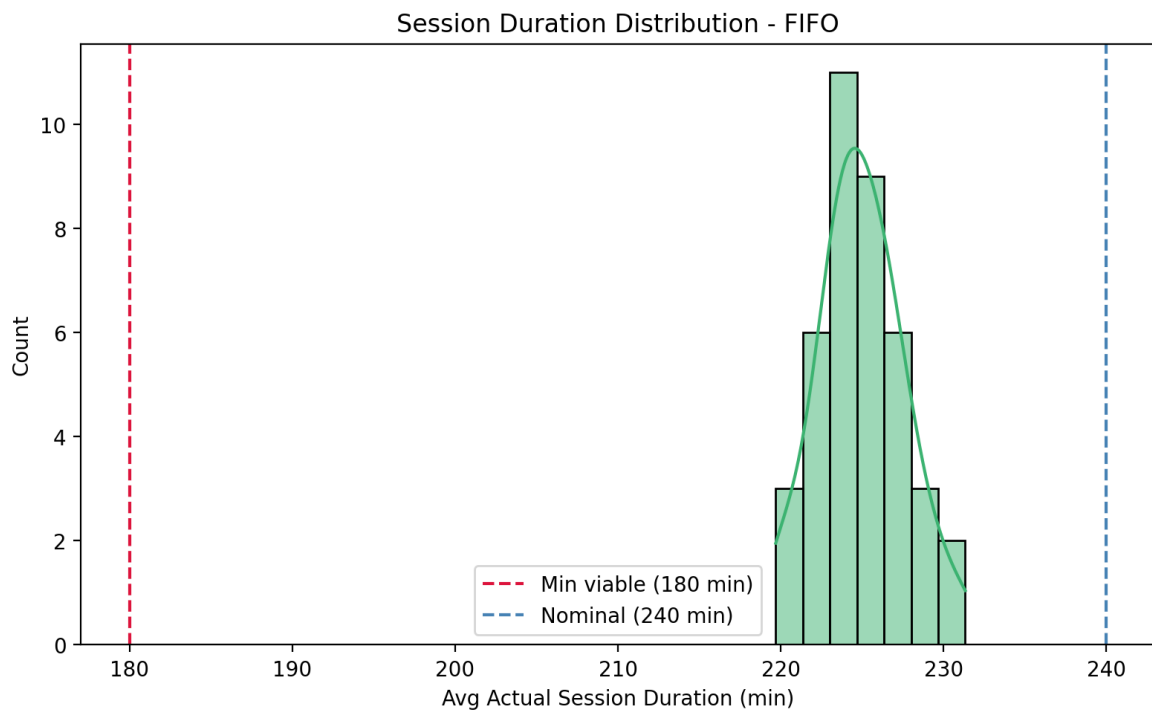


Figure 4. Actual session duration distribution (3 h min / 4 h nominal reference lines).

5. Edge-Case Stress Tests

Performance under extreme or auto-captured boundary conditions. Each card shows the full shift snapshot followed by per-strategy outcomes.

Scenario: Burst Arrival

Shift Snapshot	
Active machines	20
Defective machines	0
Total machines	20
Nurses	2
Patients	20
Arrival window	0 - 0 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results	
[PASS] FIFO Patients: 20 Failed: 0 Truncated: 4	
Wait (mean/max): 66.5 / 139.0 min Avg session: 220 min Nurse util: 41.4% Machine util: 83.6%	

Scenario: Maintenance Disaster

Shift Snapshot	
Active machines	15
Defective machines	12
Total machines	27
Nurses	4
Patients	20
Arrival window	1 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results	
[FAIL] FIFO Patients: 20 Failed: 17 Truncated: 0	
Wait (mean/max): 0.0 / 0.0 min Avg session: 215 min Nurse util: 3.1% Machine util: 63.9%	

Scenario: Marathon Shift

Shift Snapshot	
Active machines	15
Defective machines	0
Total machines	15
Nurses	2
Patients	20

Arrival window	10 - 200 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 5 | Truncated: 5

Wait (mean/max): 0.0 / 0.0 min | Avg session: 230 min | Nurse util: 41.7% | Machine util: 69.4%

Scenario: Auto: seed-0 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	18
Arrival window	2 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	0

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 5 | Truncated: 0

Wait (mean/max): 25.0 / 48.0 min | Avg session: 227 min | Nurse util: 19.6% | Machine util: 67.9%

Scenario: Auto: seed-0 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	18
Arrival window	2 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	0

Strategy Results

Scenario: Auto: seed-1 (FIXED)

Shift Snapshot

Active machines	17
Defective machines	1
Total machines	18

Nurses	4
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	1

Strategy Results

Scenario: Auto: seed-2 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	1
Total machines	16
Nurses	2
Patients	15
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	2

Strategy Results

Scenario: Auto: seed-3 (FIXED)

Shift Snapshot

Active machines	18
Defective machines	1
Total machines	19
Nurses	4
Patients	16
Arrival window	9 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	3

Strategy Results

Scenario: Auto: seed-4 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	3
Total machines	18
Nurses	3
Patients	16

Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	4

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 1 | Truncated: 0

Wait (mean/max): 28.7 / 38.0 min | Avg session: 223 min | Nurse util: 19.8% | Machine util: 66.5%

Scenario: Auto: seed-4 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	3
Total machines	18
Nurses	3
Patients	16
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	4

Strategy Results

Scenario: Auto: seed-5 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	19
Arrival window	0 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	5

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0

Wait (mean/max): 26.7 / 47.0 min | Avg session: 226 min | Nurse util: 19.8% | Machine util: 67.2%

Scenario: Auto: seed-5 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	1
Total machines	16

Nurses	3
Patients	19
Arrival window	0 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	5

Strategy Results

Scenario: Auto: seed-6 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	6

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0

Wait (mean/max): 33.7 / 65.0 min | Avg session: 231 min | Nurse util: 31.5% | Machine util: 72.2%

Scenario: Auto: seed-6 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	6

Strategy Results

Scenario: Auto: seed-7 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15

Nurses	2
Patients	17
Arrival window	2 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	7

Strategy Results

[FAIL] FIFO | Patients: 17 | Failed: 4 | Truncated: 0

Wait (mean/max): 51.7 / 82.0 min | Avg session: 220 min | Nurse util: 29.0% | Machine util: 69.9%

Scenario: Auto: seed-7 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	2
Patients	17
Arrival window	2 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	7

Strategy Results

Scenario: Auto: seed-8 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	3
Total machines	16
Nurses	3
Patients	16
Arrival window	1 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	8

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 3 | Truncated: 0

Wait (mean/max): 29.4 / 50.0 min | Avg session: 224 min | Nurse util: 17.1% | Machine util: 66.5%

Scenario: Auto: seed-8 (FIXED)

Shift Snapshot

Active machines	13
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Defective machines	3
Total machines	16
Nurses	3
Patients	16
Arrival window	1 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	8

Strategy Results

Scenario: Auto: seed-9 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	3
Total machines	17
Nurses	4
Patients	18
Arrival window	5 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	9

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 4 | Truncated: 0

Wait (mean/max): 12.1 / 34.0 min | Avg session: 226 min | Nurse util: 14.1% | Machine util: 67.6%

Scenario: Auto: seed-9 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	3
Total machines	17
Nurses	4
Patients	18
Arrival window	5 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	9

Strategy Results

Scenario: Auto: seed-10 (FIFO)

Shift Snapshot

Active machines	15
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Defective machines	5
Total machines	20
Nurses	2
Patients	19
Arrival window	0 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	10

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0

Wait (mean/max): 49.3 / 88.0 min | Avg session: 225 min | Nurse util: 31.0% | Machine util: 71.1%

Scenario: Auto: seed-10 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	5
Total machines	20
Nurses	2
Patients	19
Arrival window	0 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	10

Strategy Results

Scenario: Auto: seed-11 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	1
Total machines	15
Nurses	4
Patients	18
Arrival window	2 - 54 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	11

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 4 | Truncated: 0

Wait (mean/max): 12.6 / 51.0 min | Avg session: 223 min | Nurse util: 15.7% | Machine util: 66.6%

Scenario: Auto: seed-11 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	1
Total machines	15
Nurses	4
Patients	18
Arrival window	2 - 54 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	11

Strategy Results**Scenario: Auto: seed-13 (FIFO)****Shift Snapshot**

Active machines	15
Defective machines	2
Total machines	17
Nurses	3
Patients	17
Arrival window	4 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	13

Strategy Results

[FAIL] FIFO | Patients: 17 | Failed: 2 | Truncated: 0

Wait (mean/max): 21.3 / 41.0 min | Avg session: 225 min | Nurse util: 21.2% | Machine util: 67.3%

Scenario: Auto: seed-13 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	2
Total machines	17
Nurses	3
Patients	17
Arrival window	4 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	13

Strategy Results**Scenario: Auto: seed-14 (FIFO)**

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	4
Patients	15
Arrival window	14 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	14

Strategy Results

[FAIL] FIFO | Patients: 15 | Failed: 2 | Truncated: 0

Wait (mean/max): 12.8 / 36.0 min | Avg session: 220 min | Nurse util: 12.8% | Machine util: 65.1%

Scenario: Auto: seed-14 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	2
Total machines	15
Nurses	4
Patients	15
Arrival window	14 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	14

Strategy Results**Scenario: Auto: seed-15 (FIXED)****Shift Snapshot**

Active machines	16
Defective machines	2
Total machines	18
Nurses	2
Patients	16
Arrival window	3 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	15

Strategy Results**Scenario: Auto: seed-16 (FIXED)**

Shift Snapshot

Active machines	17
Defective machines	3
Total machines	20
Nurses	3
Patients	17
Arrival window	1 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	16

Strategy Results**Scenario: Auto: seed-17 (FIFO)****Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	3
Patients	19
Arrival window	1 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	17

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 5 | Truncated: 0

Wait (mean/max): 11.4 / 31.0 min | Avg session: 225 min | Nurse util: 19.9% | Machine util: 67.1%

Scenario: Auto: seed-17 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	3
Patients	19
Arrival window	1 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	17

Strategy Results**Scenario: Auto: seed-18 (FIFO)**

Shift Snapshot

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	18

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 2 | Truncated: 1

Wait (mean/max): 41.3 / 75.0 min | Avg session: 226 min | Nurse util: 30.7% | Machine util: 70.1%

Scenario: Auto: seed-18 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	18

Strategy Results**Scenario: Auto: seed-19 (FIFO)****Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	2
Patients	20
Arrival window	6 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	19

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 6 | Truncated: 0

Wait (mean/max): 35.4 / 70.0 min | Avg session: 226 min | Nurse util: 29.4% | Machine util: 69.1%

Scenario: Auto: seed-19 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	2
Patients	20
Arrival window	6 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	19

Strategy Results**Scenario: Auto: seed-20 (FIFO)****Shift Snapshot**

Active machines	12
Defective machines	4
Total machines	16
Nurses	4
Patients	20
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	20

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 8 | Truncated: 0

Wait (mean/max): 33.8 / 56.0 min | Avg session: 225 min | Nurse util: 11.2% | Machine util: 66.1%

Scenario: Auto: seed-20 (FIXED)**Shift Snapshot**

Active machines	12
Defective machines	4
Total machines	16
Nurses	4
Patients	20
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	20

Strategy Results

Scenario: Auto: seed-21 (FIFO)**Shift Snapshot**

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	21

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 3 | Truncated: 0

Wait (mean/max): 22.3 / 37.0 min | Avg session: 229 min | Nurse util: 16.0% | Machine util: 67.6%

Scenario: Auto: seed-21 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	21

Strategy Results**Scenario: Auto: seed-22 (FIXED)****Shift Snapshot**

Active machines	16
Defective machines	2
Total machines	18
Nurses	2
Patients	16
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	22

Strategy Results

Scenario: Auto: seed-23 (FIFO)**Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	2
Patients	17
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	23

Strategy Results

[FAIL] FIFO | Patients: 17 | Failed: 4 | Truncated: 0

Wait (mean/max): 49.1 / 75.0 min | Avg session: 225 min | Nurse util: 28.3% | Machine util: 69.8%

Scenario: Auto: seed-23 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	2
Patients	17
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	23

Strategy Results**Scenario: Auto: seed-24 (FIFO)****Shift Snapshot**

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	20
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	24

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 3 | Truncated: 0

Wait (mean/max): 28.9 / 42.0 min | Avg session: 227 min | Nurse util: 24.9% | Machine util: 71.3%

Scenario: Auto: seed-24 (FIXED)

Shift Snapshot

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	20
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	24

Strategy Results

Scenario: Auto: seed-25 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	2
Total machines	16
Nurses	2
Patients	18
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	25

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 4 | Truncated: 0

Wait (mean/max): 39.4 / 65.0 min | Avg session: 227 min | Nurse util: 31.8% | Machine util: 72.3%

Scenario: Auto: seed-25 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	2
Total machines	16
Nurses	2
Patients	18
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	25

Strategy Results

Scenario: Auto: seed-26 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	4
Total machines	17
Nurses	2
Patients	20
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	26

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 7 | Truncated: 0

Wait (mean/max): 27.2 / 53.0 min | Avg session: 224 min | Nurse util: 26.8% | Machine util: 68.8%

Scenario: Auto: seed-26 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	4
Total machines	17
Nurses	2
Patients	20
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	26

Strategy Results

Scenario: Auto: seed-27 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	20
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	27

Strategy Results**[FAIL] FIFO | Patients: 20 | Failed: 5 | Truncated: 0**

Wait (mean/max): 12.2 / 23.0 min | Avg session: 226 min | Nurse util: 18.6% | Machine util: 66.8%

Scenario: Auto: seed-27 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	20
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	27

Strategy Results**Scenario: Auto: seed-28 (FIXED)****Shift Snapshot**

Active machines	16
Defective machines	1
Total machines	17
Nurses	4
Patients	15
Arrival window	6 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	28

Strategy Results**Scenario: Auto: seed-29 (FIXED)****Shift Snapshot**

Active machines	19
Defective machines	1
Total machines	20
Nurses	2
Patients	19
Arrival window	6 - 53 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	29

Strategy Results

Scenario: Auto: seed-30 (FIFO)

Shift Snapshot

Active machines	16
Defective machines	1
Total machines	17
Nurses	3
Patients	19
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	30

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 3 | Truncated: 0

Wait (mean/max): 18.9 / 39.0 min | Avg session: 226 min | Nurse util: 22.0% | Machine util: 68.0%

Scenario: Auto: seed-30 (FIXED)

Shift Snapshot

Active machines	16
Defective machines	1
Total machines	17
Nurses	3
Patients	19
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	30

Strategy Results

Scenario: Auto: seed-31 (FIFO)

Shift Snapshot

Active machines	12
Defective machines	3
Total machines	15
Nurses	3
Patients	15
Arrival window	3 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	31

Strategy Results**[FAIL] FIFO | Patients: 15 | Failed: 3 | Truncated: 0**

Wait (mean/max): 32.2 / 48.0 min | Avg session: 220 min | Nurse util: 15.4% | Machine util: 65.2%

Scenario: Auto: seed-31 (FIXED)**Shift Snapshot**

Active machines	12
Defective machines	3
Total machines	15
Nurses	3
Patients	15
Arrival window	3 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	31

Strategy Results**Scenario: Auto: seed-32 (FIFO)****Shift Snapshot**

Active machines	14
Defective machines	1
Total machines	15
Nurses	2
Patients	15
Arrival window	0 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	32

Strategy Results**[FAIL] FIFO | Patients: 15 | Failed: 1 | Truncated: 0**

Wait (mean/max): 22.5 / 39.0 min | Avg session: 223 min | Nurse util: 27.1% | Machine util: 70.2%

Scenario: Auto: seed-32 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	1
Total machines	15
Nurses	2
Patients	15
Arrival window	0 - 57 min
Shift duration	360 min
Min viable session	180 min

Machine cooldown	60 min
Scenario seed	32

Strategy Results

Scenario: Auto: seed-33 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	19
Arrival window	3 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	33

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 5 | Truncated: 1

Wait (mean/max): 25.5 / 59.0 min | Avg session: 230 min | Nurse util: 31.5% | Machine util: 72.2%

Scenario: Auto: seed-33 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	19
Arrival window	3 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	33

Strategy Results

Scenario: Auto: seed-34 (FIFO)

Shift Snapshot

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	19
Arrival window	3 - 60 min
Shift duration	360 min
Min viable session	180 min

Machine cooldown	60 min
Scenario seed	34

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 2 | Truncated: 0**

Wait (mean/max): 20.1 / 34.0 min | Avg session: 228 min | Nurse util: 21.9% | Machine util: 68.1%

Scenario: Auto: seed-34 (FIXED)**Shift Snapshot**

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	19
Arrival window	3 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	34

Strategy Results**Scenario: Auto: seed-35 (FIFO)****Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	3
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	35

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 6 | Truncated: 0**

Wait (mean/max): 34.0 / 59.0 min | Avg session: 224 min | Nurse util: 19.2% | Machine util: 67.0%

Scenario: Auto: seed-35 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	3
Patients	19
Arrival window	0 - 55 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	35

Strategy Results

Scenario: Auto: seed-36 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	17
Arrival window	0 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	36

Strategy Results

[FAIL] FIFO | Patients: 17 | Failed: 2 | Truncated: 0

Wait (mean/max): 42.9 / 70.0 min | Avg session: 224 min | Nurse util: 32.4% | Machine util: 74.3%

Scenario: Auto: seed-36 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	17
Arrival window	0 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	36

Strategy Results

Scenario: Auto: seed-37 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	3
Total machines	16
Nurses	4
Patients	20
Arrival window	2 - 53 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	37

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 7 | Truncated: 0

Wait (mean/max): 19.7 / 46.0 min | Avg session: 224 min | Nurse util: 14.0% | Machine util: 66.8%

Scenario: Auto: seed-37 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	3
Total machines	16
Nurses	4
Patients	20
Arrival window	2 - 53 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	37

Strategy Results

Scenario: Auto: seed-38 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	4
Total machines	18
Nurses	3
Patients	20
Arrival window	0 - 49 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	38

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 6 | Truncated: 0

Wait (mean/max): 49.7 / 69.0 min | Avg session: 222 min | Nurse util: 20.6% | Machine util: 67.3%

Scenario: Auto: seed-38 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	4
Total machines	18
Nurses	3

Patients	20
Arrival window	0 - 49 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	38

Strategy Results

Scenario: Auto: seed-39 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	2
Total machines	16
Nurses	3
Patients	16
Arrival window	0 - 52 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	39

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 2 | Truncated: 0

Wait (mean/max): 34.7 / 45.0 min | Avg session: 229 min | Nurse util: 17.1% | Machine util: 67.4%

Scenario: Auto: seed-39 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	2
Total machines	16
Nurses	3
Patients	16
Arrival window	0 - 52 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	39

Strategy Results